

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt
```

```
data=pd.read_csv('/content/Weight_prediction.csv')
```

```
data.head()
```

	Weight	Height	Age	Gender
0	79	1.80	35	Male
1	69	1.68	39	Male
2	73	1.82	25	Male
3	95	1.70	60	Male
4	82	1.87	27	Male

```
data.shape
```

```
(10, 4)
```

```
data.isnull().sum()
```

```

0
Weight  0
Height  0
Age     0
Gender  0

dtype: int64
```

```
data['Gender'] = data['Gender'].replace('Ma le', 'Male')
data['Gender'] = data['Gender'].str.strip()

data_encoded = pd.get_dummies(data, columns=['Gender'], drop_first=True, dtype=int)

x = data_encoded[['Height', 'Age', 'Gender_Male']].values
y = data_encoded[['Weight']].values
```

Start coding or [generate](#) with AI.

```
[[79]
 [69]
 [73]
 [95]
 [82]
 [55]
 [69]
 [71]
 [64]
 [69]]
[[1.8 ]
 [1.68]
 [1.82]
 [1.7 ]
 [1.87]
 [1.55]
 [1.5 ]
 [1.78]
 [1.67]
 [1.64]]
```

```
x_train, x_test, y_train, y_test = train_test_split(x,y,test_size=0.2, random_state = 42)
```

```
model=LinearRegression()
model.fit(x_train, y_train)
```

```
y_pred=model.predict(x_test)
```

```
print(y_pred)
print(y_test)
```

```
[[56.85651025]
 [79.10576115]]
[[64]
 [69]]
```

```
import numpy as np

plt.scatter(x_train[:, 0], y_train, color='blue')

height_min, height_max = x_train[:, 0].min(), x_train[:, 0].max()
height_range = np.linspace(height_min, height_max, 100).reshape(-1, 1)

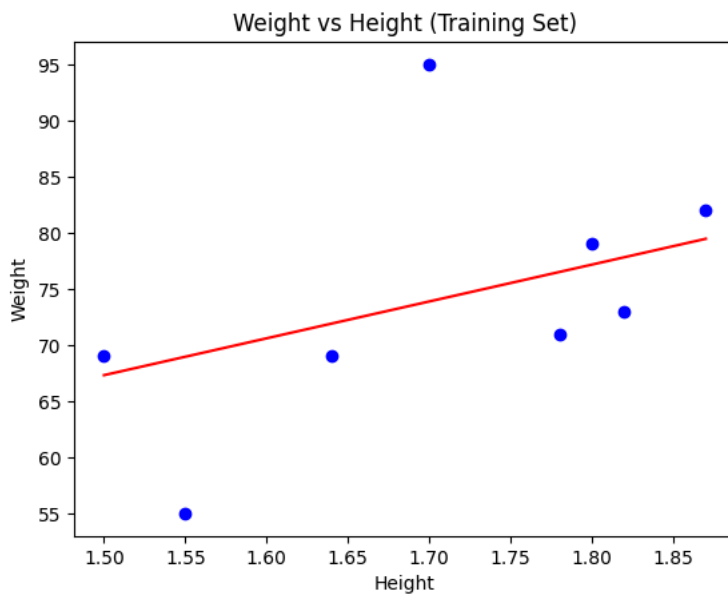
mean_age = x_train[:, 1].mean()
mean_gender_male = x_train[:, 2].mean()

dummy_x_for_plot = np.hstack([height_range,
                               np.full(height_range.shape, mean_age),
                               np.full(height_range.shape, mean_gender_male)])

predicted_weights_for_plot = model.predict(dummy_x_for_plot)

plt.plot(height_range, predicted_weights_for_plot, color='red')

plt.title('Weight vs Height (Training Set)')
plt.xlabel('Height')
plt.ylabel('Weight')
plt.show()
```



```
import numpy as np

plt.scatter(x_test[:, 0], y_test, color='blue')

height_min_test, height_max_test = x_test[:, 0].min(), x_test[:, 0].max()
height_range_test = np.linspace(height_min_test, height_max_test, 100).reshape(-1, 1)

dummy_x_for_plot_test = np.hstack([height_range_test,
                                    np.full(height_range_test.shape, mean_age),
                                    np.full(height_range_test.shape, mean_gender_male)])

predicted_weights_for_plot_test = model.predict(dummy_x_for_plot_test)

plt.plot(height_range_test, predicted_weights_for_plot_test, color='red')

plt.title('Weight vs Height (Testing Set)')
plt.xlabel('Height')
```

```
plt.ylabel('Weight')  
plt.show()
```

